

LAB TASK#3

NAME: MASHAAL ISHTIAQ

ROLL NO: 22P-9326

SECTION: BCS-2C

OUTPUT:

```
#include<iostream>
using namespace std;

class List{

public:
//publically declaring data members as we dont want to use getter setters
int *arr;
int length;
int capacity;
int *curr=arr;

//constructor
List(int size){
//data members initialization
int*curr;
arr=new int[size];
capacity=size;
length=0;
curr=arr;
cout << "Enter the elements of array (Enter -1 to stop): ";
int input=0;

for(int i=0;i<size;i++){
if(input==-1){
break;
}
cin>>input;
this->arr[i]=input;
this->length++;

}
}

//destructor
~List(){
delete[] arr;
}

void next(){
curr++;
}

void back(){
curr--;
}
```

```
void start(){
curr=arr;
}

void tail(){
curr=arr+length-1;
}
//copy constructor
List(const List& other){
capacity=other.capacity;
length=other.length;
arr=new int[capacity];
for(int i=0;i<length;i++){
arr[i]=other.arr[i];
}
curr=arr+(other.curr-other.arr);
cout<<"list copied"<<endl;
}


void Clear(){
length=0;
}

void Insert(int val){

isFull();
if(length<capacity){
tail();
*(curr+1)=val;
if(length==0){
*curr=val;
}
length++;
}
}

void Insert_to(int val,int pos){
isFull();
if(length<capacity){
if(pos<1 || pos>length+1){
cout<<"Invalid position"<<endl;
return;
}}

tail(); //pointer will reach to the tail value

for(int i=length;i>=pos;i--){

*(curr+1)=*curr;
back();
}
```

```
*(curr+1)=val;
length++;
}

void Print_list(){

for(int i=0;i<length;i++){
cout<<*(arr+i)<<" ";
}
cout<<endl;
}
void isFull(){
if(length==capacity){
cout<<"Array is full"<<endl;
}
else
cout<<"array is not full"<<endl;

}

void isEmpty(){
if(length==0){
cout<<"Array is empty"<<endl;
}
else{
cout<<"array not empty"<<endl;
}
}

void Remove(int val){
isEmpty();
if(length>0){
int pos=Find(val);

for(int i=pos;i<=length;i++){
*(curr)=*(curr+1);
next();
}
length--;
}}

void RemoveTo(int pos){
isEmpty();
if(length>0){
if(pos<1 || pos>length){
cout<<"Invalid position"<<endl;
return;
}
curr=arr+pos-1;
for(int i=pos;i<=length;i++){
*(curr)=*(curr+1);
next();
```

```
}
length--;
}}

int Get(int pos){
isEmpty();
if(length>0){
if(pos<1 || pos>length){
cout<<"Invalid position"<<endl;
return 0;
}
curr=arr+pos-1;
}
return *curr;
}

void Update(int pos,int val){

isEmpty();
if(length>0){
if(pos<1 || pos>length){
cout<<"Invalid position"<<endl;
return;
}
curr=arr+pos-1;
*curr=val;

}}

int Find(int val){
isEmpty();
if(length>0){
start();

for(int i=1;i<=length;i++){
if(*curr==val){
return i;
}
next();
}}
cout<<"Value not found"<<endl;
return -1;
}

void Reverse_list(){

isEmpty();
if(length>0){

int left=0;
int right=left+length-1;
```

```
while(left<right){

int temp=arr[left];
arr[left]=arr[right];
arr[right]=temp;
left++;
right--;
}

}}

void Sort_list(){

isEmpty();
if(length>0){
for( int i=0;i<length;i++){
for(int j=i+1;j<length;j++){
if(arr[j]<arr[i] ){
// Swap elements
int temp=arr[i];
arr[i]=arr[j];
arr[j]=temp;
}

}
}

}
}

};

int main(){

int size,found,value;
int val;
int pos;
char ch;

cout<<"Enter size of the array: "<<endl;
cin>>size;
while(1){
cout<<"___Enter the operation you want to perform___"<<endl;
cout<<"1. create list"<<endl<<"2. copy"<<endl<<"3. clear"<<endl<<"4. insert"<<endl<<"5.
remove"<<endl<<"6. get"<<endl<<"7. update"<<endl<<"8. find"<<endl<<"9.
length"<<endl<<"10. print List"<<endl<<"11. reverse list"<<endl<<"12. sort in ascending
order"<<endl<<"13. isFull()"<<endl<<"14. isEmpty"<<endl<<"15. none"<<endl;
int choice;
cout<<"Enter your choice from above: "<<endl;
cin>>choice;

List *list;
```

```
List *list2;
switch(choice){
case 1:
list=new List(size); // constructor called
break;

case 2:
list2=new List(*list); //copy constructor called->deep copy
delete list2;
break;

case 3:
list->Clear();
cout<<"list cleared"<<endl;
break;

case 4:
cout<<"Select which insert function you want to perform"<<endl;
cout<<"a. intsert a value"<<endl<<"b. insert at a certain position"<<endl;
cin>>ch;

if(ch=='a'){
cout<<"Enter the value you want to insert"<<endl;
cin>>val;
list->Insert(val);
}
if(ch=='b'){
cout<<"Enter the value you want to insert"<<endl;
cin>>val;

cout<<"Enter the position where you want to insert"<<endl;
cin>>pos;

list->Insert_to(val,pos);

}
else{
cout<<"Invalid selection"<<endl;
}
break;

case 5:
cout<<"Select the remove function you want to perform" <<endl;
cout<<"a. Remove value"<<endl<<"b. Remove value on certain position"<<endl;
cin>>ch;
if(ch=='a'){
cout<<"Enter the value you want to remove"<<endl;
cin>>val;
list->Remove(val);
cout<<"value removed successfully"<<endl;
}
if(ch=='b'){
```

```
cout<<"Enter the position where you want to remove the value from"<<endl;
cin>>pos;
list->RemoveTo(pos);
cout<<"value removed successfully from the given position"<<endl;

}
break;
case 6:
cout<<"Enter the position from which you want to retrieve value"<<endl;
cin>>pos;
value=list->Get(pos);
cout<<"Value: "<<value<<endl;
break;

case 7:
cout<<"Enter the position where you want to enter the new value: "<<endl;
cin>>pos;
cout<<"Enter the value now: "<<endl;
cin>>val;

list->Update(pos,val);
break;

case 8:
cout<<"Enter the value you want to find"<<endl;
cin>>val;

found=list->Find(val);
cout<<"value found at index: "<<found<<endl;
break;

case 9:
list->isEmpty();
cout<<"length is: "<<list->length<<endl;
break;

case 10:
list->Print_list();
break;

case 11:
list->Reverse_list();
cout<<"List successfully reversed"<<endl;
list->Print_list();
break;
case 12:
list→Sort_list();



cout<<"array sorted in ascending order"<<endl;
list->Print_list();
```

```
break;
case 13:
list->isFull();
break;

case 14:
list->isEmpty();
break;

case 15:
break;

default:
cout<<"Invalid output"<<endl;
}
}

}
```

```
15. none
Enter your choice from above:
2
list copied
___Enter the operation you want to perform___
```

```
Enter size of the array:
4
___Enter the operation you want to perform___
1. create list
2. copy
3. clear
4. insert
5. remove
6. get
7. update
8. find
9. length
10. print List
11. reverse list
12. sort in ascending order
13. isFull()
14. isEmpty
15. none
Enter your choice from above:
1
Enter the elements of array (Enter -1 to stop): 9
8
7
```

```
6
___Enter the operation you want to perform___
1. create list
2. copy
3. clear
4. insert
5. remove
6. get
7. update
8. find
9. length
10. print List
11. reverse list
12. sort in ascending order
13. isFull()
14. isEmpty
15. none
Enter your choice from above:
14
array not empty
___Enter the operation you want to perform___
1. create list
2. copy
```

```
Enter your choice from above:
13
Array is full
___Enter the operation you want to perform___
```

```
Enter your choice from above:
14
array not empty
___Enter the operation you want to perform___
```

```
Enter your choice from above:
12
array not empty
array sorted in ascending order6 7 8 9
```

```
Enter your choice from above:
11
array not empty
List successfully reversed
```

```
Enter your choice from above:
8
Enter the value you want to find
2
array not empty
value found at index: 2
```

```
15. none
Enter your choice from above:
6
Enter the position from which you want to
2
array not empty
```

```
15. none
Enter your choice from above:
5
Select the remove function you want to perform
a. Remove value
b. Remove value on certain position
b
Enter the position where you want to remove the value from
1
array not empty
value removed successfully from the given position
___Enter the operation you want to perform___
```

```
14. isEmpty
15. none
Enter your choice from above:
10
3 4
___Enter the operation you want to perform___
1. create list
2. copy
```

```
Enter your choice from above:
4
Select which insert function you want to perform
a. intsert a value
b. insert at a certain position
a
Enter the value you want to insert
5
array is not full
Invalid selection
```

```
4
Select which insert function you want to perform
a. intsert a value
b. insert at a certain position
b
Enter the value you want to insert
9
Enter the position where you want to insert
1
array is not full
___Enter the operation you want to perform___
```